

230905232
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Roll No. 37

WPL Week 4

Lab Assignments

1. Write a python program to reverse a content a file and store it in another file.

Code:

```
with open('input.txt', 'r') as file:  
    contents = file.read()
```

```
with open('output.txt', 'w+') as file:  
    file.write(contents[::-1])
```

input file (input.txt):

this is a test file, the outputs will be reversed

tuptuo eht ni elbadaer si trap siht

output file (output.txt):

this part is readable in the output

desrever eb lliw stuptuo eht ,elif tset a si siht

```
● WPC2@selab-05:~/230905232/WP-Lab/lab04-python-basics$ python3 1.py
```

```
lab04-python-basics > ≡ output.txt
```

```
1  this part is readable in the output
```

```
2
```

```
3  desrever eb lliw stuptuo eht ,elif tset a si siht
```

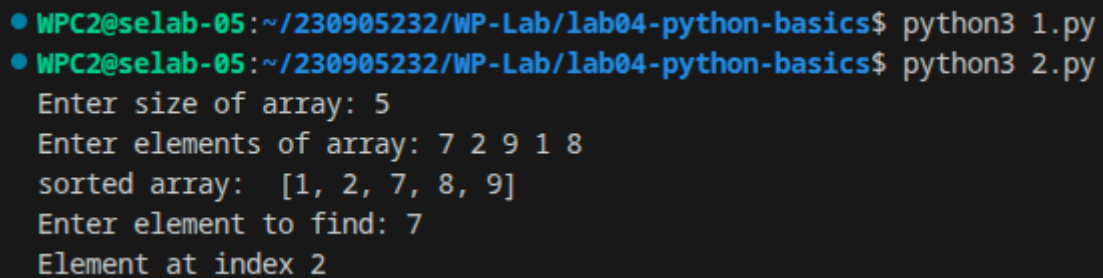
2. Write a python program to implement binary search with recursion.

Code:

```
def bin_search(arr, l, r, key):
    if l > r: return -1
    n = (l+r)//2
    if arr[n] > key:
        return bin_search(arr, l, n-1, key)
    elif arr[n] < key:
        return bin_search(arr, n+1, r, key)
    else:
        return n

n = int(input("Enter size of array: "))
arr = [0] * n
arr = sorted(list(map(int, input("Enter elements of array: ").split()))))
print("sorted array: ", arr)
key = int(input("Enter element to find: "))
res = bin_search(arr, 0, len(arr)-1, key)
if res == -1:
    print("Element not in array")
else:
    print("Element at index", res)
```

output:



```
● WPC2@selab-05:~/230905232/WP-Lab/lab04-python-basics$ python3 1.py
● WPC2@selab-05:~/230905232/WP-Lab/lab04-python-basics$ python3 2.py
Enter size of array: 5
Enter elements of array: 7 2 9 1 8
sorted array:  [1, 2, 7, 8, 9]
Enter element to find: 7
Element at index 2
```

3. Write a python program to sort words in alphabetical order.

Code:

```
def merge(l, r):
    res = []
    i = j = 0

    while i < len(l) and j < len(r):
        if l[i] < r[j]:
            res.append(l[i])
            i += 1
        else:
            res.append(r[j])
            j += 1

    res.extend(l[i:])
    res.extend(r[j:])
    return res

def sort_words(arr):
    n = len(arr)
    if n <= 1:
        return arr
    if n == 2:
        if arr[0] > arr[1]:
            arr[0], arr[1] = arr[1], arr[0]
        return arr

    left = sort_words(arr[:n//2])
    right = sort_words(arr[n//2:])

    return merge(left, right)

print("Enter words to sort:")
strings = list(input().split())
strings = sort_words(strings)
print("Sorted words: ", strings)
```

output:

```
WPC2@selab-05:~/230905232/WP-Lab/lab04-python-basics$ python3 3.py
Enter words to sort:
this is a test
Sorted words:  ['a', 'is', 'test', 'this']
```

4. Write a Python class to get all possible unique subsets from a set of distinct integers

Input:[4,5,6]

Output : [[], [6], [5], [5, 6], [4], [4, 6], [4, 5], [4, 5, 6]]

code:

```
def subsets(arr):
    res = []

    def backtrack(curr, i):
        if i == len(arr):
            res.append(curr.copy())
            return

        backtrack(curr, i+1)

        curr.append(arr[i])
        backtrack(curr, i+1)
        curr.pop()

    backtrack([], 0)
    return res

arr = list(map(int, input("Enter elements of the array to generate all possible subsets:\n").split()))
print(subsets(arr))
```

output:

```
● WPC2@selab-05:~/230905232/WP-Lab/lab04-python-basics$ python3 4.py
Enter elements of the array to generate all possible subsets:
4 5 6
[[], [6], [5], [5, 6], [4], [4, 6], [4, 5], [4, 5, 6]]
```

5. Write a Python class to find a pair of elements (indices of the two numbers) from a given array whose sum equals a specific target number.

Input: numbers= [10,20,10,40,50,60,70], target=50

Output: 2, 3.

code:

```
class TwoSum:
```

```
    def __init__(self, arr, target):
```

```
        self.arr = arr
```

```
        self.target = target
```

```
    def find_pair(self):
```

```
        seen = {}
```

```
        for idx, num in enumerate(self.arr):
```

```
            n = self.target - num
```

```
            if n in seen:
```

```
                return (seen[n], idx)
```

```
            seen[num] = idx
```

```
        return None
```

```
ts = TwoSum([], 0)
```

```
ts.arr = list(map(int, input("Enter elements of the array: ").split()))
```

```
ts.target = int(input("Enter target: "))
```

```
print("Elements at indices", ts.find_pair())
```

output:

```
WPC2@selab-05:~/230905232/WP-Lab/lab04-python-basics$ python3 5.py
Enter elements of the array: 10 20 10 40 50 60 70
Enter target: 50
Elements at indices (2, 3)
```

6. Write a Python class to implement pow(x, n).

code:

```
import math
```

```
class Power:
```

```
    @staticmethod
```

```
    def pow(x, n):
```

```
        if n == 0: return 1
```

```
        if x == 0 or x == 1: return int(x)
```

```
        if n == 1: return x if not float(x).is_integer() else int(x)
```

```
        n_neg_flag = 0
```

```
        x_neg_flag = 0
```

```
        original_n = n
```

```
        original_x = x
```

```
        if n == 0: return 1
```

```
        if n < 0:
```

```
            n *= -1
```

```
            n_neg_flag = 1
```

```
        if x < 0:
```

```
            x_neg_flag = n%2
```

```
            x = -x
```

```
        res = math.exp(n * math.log(x))
```

```
        if n_neg_flag: res = 1/res
```

```
        if x_neg_flag: res *= -1
```

```
        if float(original_n).is_integer() and float(original_x).is_integer() and not n_neg_flag:
```

```
            return int(round(res, 0))
```

```
        return res
```

```
x, n = list(map(float, input("Enter x and n: ").split()))
```

```
res = Power.pow(x,n)
```

```
if float(n).is_integer() and float(x).is_integer():
```

```
    x, n = int(x), int(n)
```

```
print(f"pow({x}, {n}) = {round(res, 6)}")
```

output:

```
● WPC2@selab-05:~/230905232/WP-Lab/lab04-python-basics$ python3 6.py
Enter x and n: 2 10
pow(2, 10) = 1024
```

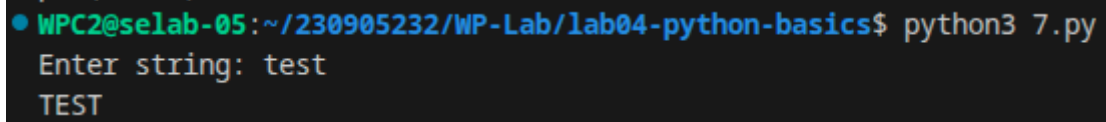
7. Write a Python class which has two methods `get_String` and `print_String`. The `get_String` accept a string from the user and `print_String` print the string in upper case.

Code:

```
class String:
    def get_String(self):
        self.str = input("Enter string: ")
    def print_String(self):
        print(self.str.upper())
```

```
s = String()
s.get_String()
s.print_String()
```

output:

A terminal window with a black background and green text. The prompt is 'WPC2@selab-05:~/230905232/WP-Lab/lab04-python-basics\$'. The command 'python3 7.py' has been executed. The program prompts 'Enter string: test' and then outputs 'TEST' in all caps.

```
WPC2@selab-05:~/230905232/WP-Lab/lab04-python-basics$ python3 7.py
Enter string: test
TEST
```